

LAB EXPERIMENT RECORD

SET 6 Product Sales Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
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AIM

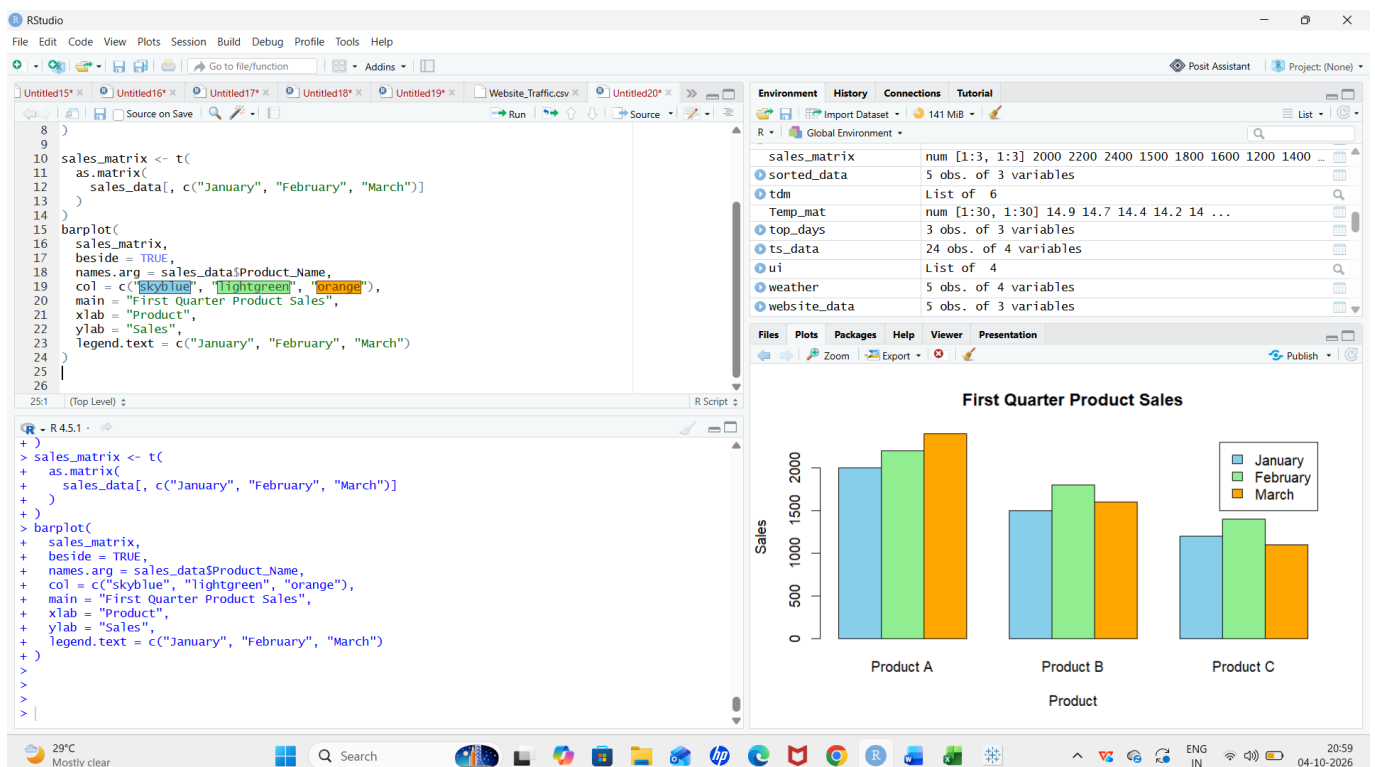
To analyze monthly product sales data using R programming by creating a grouped bar chart, stacked area chart, and table, and to develop a Tableau dashboard for interactive exploration of product sales data.

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter.

```
# Monthly Product Sales Dataset

sales_data <- data.frame(
  Product_Name = c("Product A", "Product B", "Product C"),
  January = c(2000, 1500, 1200),
  February = c(2200, 1800, 1400),
  March = c(2400, 1600, 1100)
)
sales_matrix <- t(
  as.matrix(
    sales_data[, c("January", "February", "March")]
  )
)
barplot(
  sales_matrix,
  beside = TRUE,
  names.arg = sales_data$Product_Name,
  col = c("skyblue", "lightgreen", "orange"),
  main = "First Quarter Product Sales",
  xlab = "Product",
  ylab = "Sales",
  legend.text = c("January", "February", "March")
)
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three

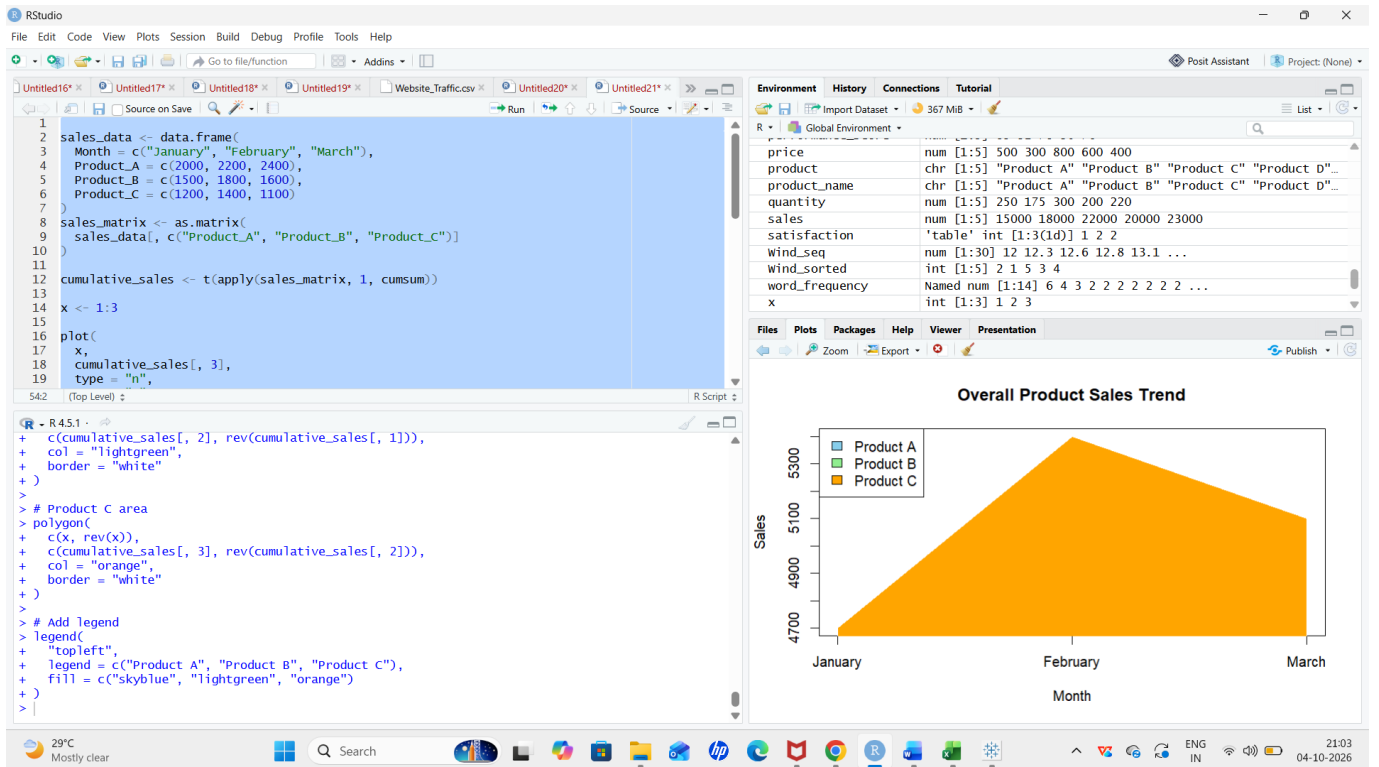
```
sales_data <- data.frame(
  Month = c("January", "February", "March"),
  Product_A = c(2000, 2200, 2400),
  Product_B = c(1500, 1800, 1600),
  Product_C = c(1200, 1400, 1100)
)
sales_matrix <- as.matrix(
  sales_data[, c("Product_A", "Product_B", "Product_C")]
)

cumulative_sales <- t(apply(sales_matrix, 1, cumsum))

x <- 1:3

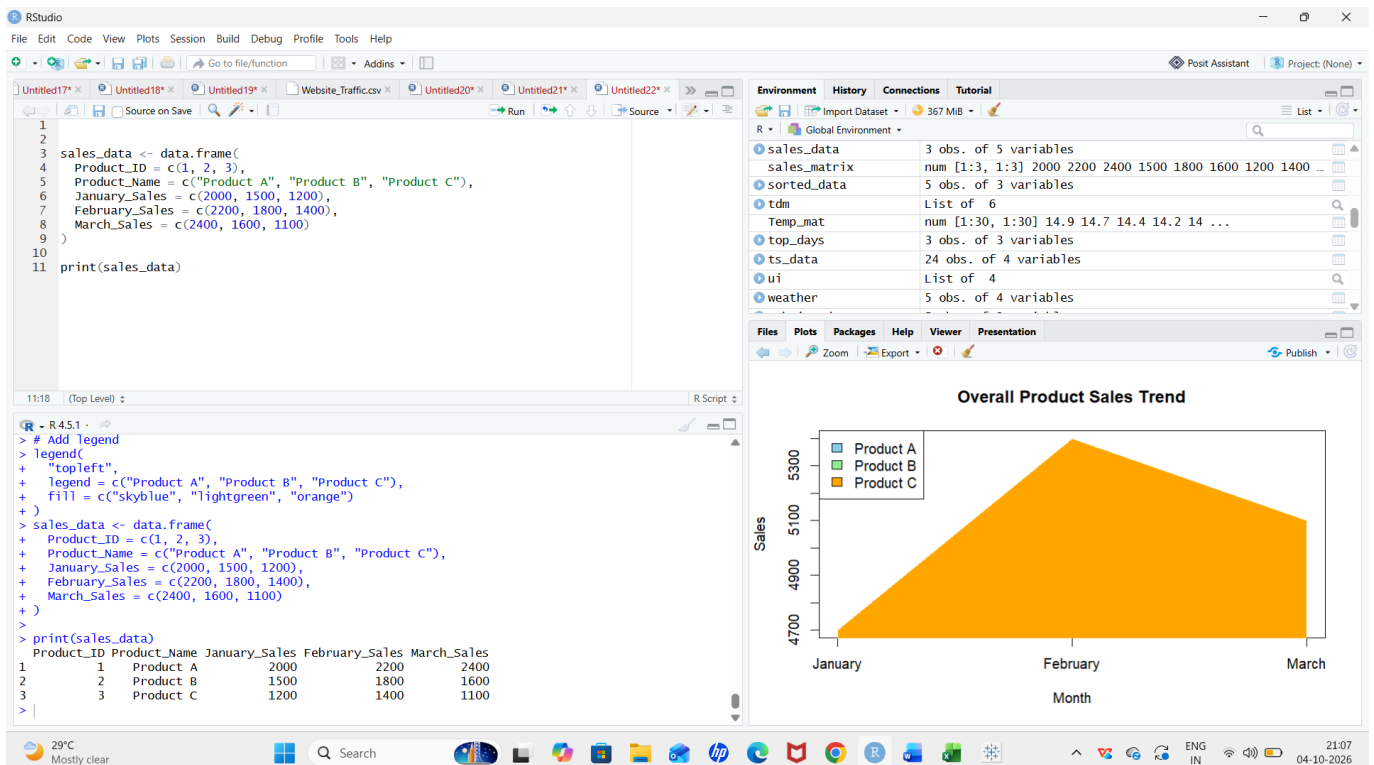
plot(
  x,
  cumulative_sales[, 3],
  type = "n",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales",
  main = "Overall Product Sales Trend"
)
axis(
  1,
  at = x,
  labels = sales_data$Month
)

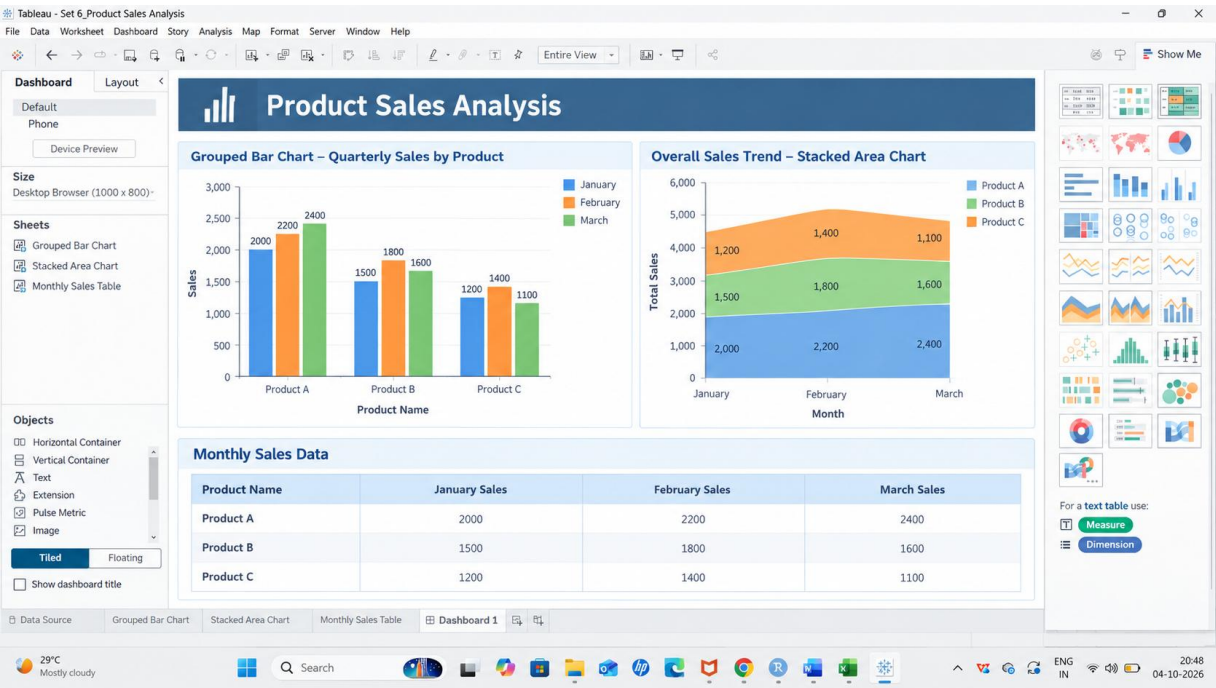
polygon(
  c(x, rev(x)),
  c(cumulative_sales[, 1], rev(rep(0, 3))),
  col = "skyblue",
  border = "white"
)
```



3. Build a table to show the monthly sales data for each product.

```
sales_data <- data.frame(  
  Product_ID = c(1, 2, 3),  
  Product_Name = c("Product A", "Product B", "Product C"),  
  January_Sales = c(2000, 1500, 1200),  
  February_Sales = c(2200, 1800, 1400),  
  March_Sales = c(2400, 1600, 1100)  
)  
  
print(sales_data)
```





PROCEDURE

1. Create the **Monthly Product Sales** dataset in R with Product ID, Product Name, January Sales, February Sales, and March Sales.
2. Use the appropriate R plotting function to create a **grouped bar chart** showing the sales of each product during the first quarter.
3. Add suitable **axis labels, title, legend, and chart elements** to the grouped bar chart.
4. Arrange the monthly sales data for all products to represent the sales trend over January, February, and March.
5. Create a **stacked area chart** to visualize the overall sales trend of all products across the three months.
6. Add appropriate **title, axis labels, and legend** to the stacked area chart.
7. Create a **table** displaying the monthly sales data for each product with proper column headings.
8. Import the product sales dataset into **Tableau**.
9. Create a **grouped bar chart** in Tableau to compare January, February, and March sales for each product.
10. Create a **stacked area chart** in Tableau to represent the overall sales trend of all products.
11. Create a **table** in Tableau to display the monthly sales data for each product.
12. Combine the grouped bar chart, stacked area chart, and table into a **Tableau dashboard**.
13. Add filters and interactive features to allow users to explore the product sales data.

RESULT

The monthly product sales data was successfully analyzed using **R programming**. The grouped bar chart displayed and compared the first-quarter sales of each product, while the stacked area chart showed the overall sales trend across January, February, and March. A table was created to clearly display the monthly sales data for each product. Finally, a **Tableau dashboard** was developed by combining the grouped bar chart, stacked area chart, and table, allowing users to interactively explore the product sales data.

